LEF MODEL · CHAPTER 28

Chapter 28: The Research Map of the LEF Model (Part B) — the Pacific, Mars, the Solar System and the Key Tests (Topics 80–132)

Based on Paper 14 — 14. Research topics: What the model opens for other researchers (Divisions H–L and the follow-on fields: topics 80–132, the three selection lists and the summary of the series)

01 · THE MYSTERY

The mystery: how does the research map continue from the floor of the Pacific to Mars, the solar system and the two key observations?

In Chapter 27 we surveyed the first 79 topics of Paper 14 (Divisions A–G), which dealt with the early Earth, the formation of the Moon and the South Pole–Aitken basin. In this closing chapter we will complete **the remaining 53 topics (Divisions H–K, topics 80–132)**, organise all 132 topics by **the three selection lists of Part L**, and close with the two overarching fields marked for follow-up outside the list:

1. **Division H — the stripes and the Pacific (topics 80–90):** the 9 studies of the parallel stripes on the floor of the North Pacific (from Mendocino to Clipperton), comparing the Pacific LLVP lower anomaly with the African one, and a blind reconstruction of the Pacific lithosphere.
2. **Divisions I and J — Mars and the whole solar system (topics 91–120):** the launch of Phobos and Deimos from Pavonis Mons and the decision of the

MMX mission (landing in 2027 and sample return in 2031), alongside the arc of moon births from Mercury to Pluto and Haumea.

3. **Division K and Part L — the key requirements and the three working lists (topics 121–132):** uniting the two decisive tests of the whole model — **the concentric exposure geometry (topics 121–122)** and **the origin of the craters in returning terrestrial material (topics 49 and 123)**.

02 · AN IMAGE

An image: the folds of a blanket in the Pacific, a steam cannon on Mars, and two litmus tests that decide the theory

Four tangible images gather the last 53 topics of Paper 14 and the decision strategy of the series:

- **A blanket stretched forward and crumpled behind (Division H, topics 80–90):** the Earth's pear, whose base stood near Kiritimati (1.9°N, 157.5°W) and which leaned west against the Earth's rotation, stretched and folded its front side into parallel stripes east of the base, crumpled its back into the western arcs of the Pacific (8,100–10,300 km to the west), and left on the floor, in a mirror image enlarged 1.37 times, the imprint of the maria (topic 87)!
- **A gas cannon at a pressure of 0.25–0.5 GPa with a barrel 200–300 km long (Division I, topics 91–106):** young Mars (radius 1.5×, a day of 5 hours) launched one after the other two lithospheric corks — first the small, wet Deimos and then the dry, hard Phobos — from a single ejection conduit at Pavonis Mons (1.48°N), in which Coriolis friction on the western wall carved the grooves of Phobos (topic 94) and a steam ring circularised their orbits!
- **Two litmus tests that decide the whole model (Part L.1):** from the 132 topics, the author distils **two key observations**: (1) whether the crust and the craters are arranged in three concentric belts around the centre of the near side (0°–31°–70°–180°, topics 121–122); and (2) whether the nucleosynthetic isotopes of the metal in the Chang'e-6 norites and in the crater melt (Cr, Ti, Mo, Ru, Os) are purely terrestrial and not asteroidal (topics 49 and 123)!

- **The clock of the axis system and the magnetic map of the SPA basin (the follow-on fields outside the list):** beyond the 132 numbered topics stand two long-term overarching projects — solving the evolution of the four axis planes of the Earth–Moon system (from an initial tilt of 3.8° to 23.44°), and deciphering the ancient magnetism of the Moon, which was erased at the centre of SPA and strengthened at its edges.

03 · SCIENCE AND MODEL

Science and model: summary of Divisions H–K and the three selection lists of Part L (topics 80–132)

Divisions H–K close the geophysical circle from the depths to the ends of the solar system and set all 18 requirements of Paper 13 as defined research topics:

1. **Division H — the stripes and the Pacific (topics 80–90):** fitting the stripes' equator, tilted by 11°–20° (topic 80), and a Monte Carlo test of the closeness of the base at Kiritimati (1.6°, topic 81); the one-sidedness of the stripes east of the base (topic 82); steps of depth and age (topic 83); the asymmetry of the western arcs (topic 84); the continuity of the stripes beneath the ridges and the shortening of the ridges at four intersections (topic 85); distinguishing "spatula" ridges drawn from above (Line, Marshall and Hawaii) from plume ridges from below (topic 86); the mirror image of the maria at a scale of 1.37× (topic 87); the continuation of the fold beneath crustal layers younger than 180 Myr in DSDP/IODP boreholes (topic 88); comparing the Pacific LLVP with the African one, and the isotopes of the plumes (He, W, Ne, Xe — topic 89); and a blind palaeomagnetic reconstruction of the lithosphere (topic 90).

2. **Division I — Mars and other bodies (topics 91–106):** the shrinking of young Mars and the loss of 92% of its rotation through the escape of 5%–20% of its mass (topic 91); the gas push in a conduit 200–300 km long at a pressure of 0.25–0.5 GPa (topic 92); the cycle of the viscous Vulcanian cork (topic 93); Coriolis friction ($2\omega v \approx 1.8$–2.0^2 m/s) and the braking of Phobos to 2.4–2.6 km/s (topic 94); the condensed steam ring (~ 10^{16} kg, topic 95); the tidal climb of Deimos to 23,463 km with dissipation boosted 8–20 times (topic

96); the race of Phobos with the corotation line, which met at ~ 19,000 km at a day of ~ 22 h (topic 97); analysis of the samples of the **MMX mission (landing 2027, return 2031)** to test isotopes, gas bubbles and hydrated minerals (topic 98); grooves and pit chains on Phobos (topic 99); the depth of the hollows on Deimos (1.9%) against Phobos (5.5%, topic 100); the centre-of-mass offset of Phobos toward Mars (topic 101); the deep caldera of Pavonis Mons (47 km wide, ~ 5 km deep, a volume of 8,300–8,700 km^3 matching the volume of the two moons, 6,740 km^3) and the mass deficit beneath it (topic 102); an ancient 4 Gyr dust component rich in solar-wind gases (topic 103); the water beneath Tharsis (topic 104); Venus, Mercury and fast super-Earths (topic 105); and the height difference between the poles as a function of the rotation rate at birth (topic 106).

3. **Division J — moon births in the solar system (topics 107–120):** the arc of births from the Sun with a peak at Saturn (2.5×10^{-4}, topic 107); the absence of a bulge at Mercury and Venus (topic 108); calculating the water that escaped through the corotation line from Uranus and Neptune (at least 2%–3%, topic 109); the density of the giants' inner bodies without the gas envelope (topic 110); the density order of Saturn's medium moons (topic 111) and of Uranus' moons (topic 112); Uranus' 98° tilt (topic 113); grooves as the mark of a forced exit through a narrow opening on Phobos and Miranda (topic 114); the tearing scars of Pluto and Charon (topic 115) and the time of ejection of its four small moons (topic 116); the density rule in split pairs — the larger partner is denser (topic 117); the capture rule at the boundary for Nereid and Himalia (2 points, topic 118); Haumea (on the verge of splitting at 3.9 h) and Ceres (2.54 R, topic 119); and the equatorial ridge of Iapetus, Pan and Atlas as spatula ridges at the time of separation (topic 120).

04 · THE MECHANISM IN WORDS

The mechanism in words: the three working lists of Part L and the two follow-on fields outside the list

Part L and the future-work section of Paper 14 gather all 132 topics into a practical plan of action for researchers according to the working tools they have:

1. **List 1 — the two decisive key observations (Part L.1):**
    - 2. **(a) The concentric exposure geometry (topics 121 and 122):** testing whether the crustal thickness and the ancient crater population are arranged in three concentric belts around the contact centre on the near side (0°–31° the contact dome, 31°–70° the edge belt, 70°–180° the far side). Since topics 121 and 122 appear *also in the public-data list*, the most decisive study for testing the model is also the most accessible for immediate execution.
    - 3. **(b) The origin of the craters in returning terrestrial material (topics 49 and 123):** determining whether ~ 99% of the Moon's craters were formed by returning terrestrial material and not by outside asteroids (which would have left 4.4–5.0 × 10^{19} kg, about 2%–6% foreign mass in the top 5 km of the crust, against only 0.9%–2% actually measured) — by measuring nucleosynthetic isotopes (Cr, Ti, Mo, Ru, Os) in the Chang'e-6 norites and in the melt of the maria.
4. **List 2 against List 3 — dividing the 132 topics by readiness (Parts L.2–L.3):** of the 132 topics in the series, **55 topics** can be carried out at once on existing public data (including all 9 studies of the Pacific stripes 80–88, 13 of the 14 studies of the maria 31–43, and the key topics 121–122); **57 topics** require the development of analytical models or heavy simulations (first among them a three-dimensional SPH simulation of the attached body — topic 12 — and a coupled thermal-orbital integration — topic 24); and **20 topics** require space missions, samples or new laboratory experiments (first among them the MMX mission to Phobos and high-pressure experiments at 50–150 GPa).
5. **The two large fields of research outside the list (Further Work):** (1) **the Earth–Moon axis system** — following the development of the four dynamic planes (the Earth's equator, the Moon's orbit, the lunar equator and the ecliptic plane) from an initial tilt of 3.8° to 23.44°, and testing whether true polar wander moved the SPA basin relative to the Earth's burner; and (2) **the**

magnetism of the Moon — deciphering the ancient magnetic field that was erased at the centre of the SPA basin and strengthened at its edges (including the swirls of Mare Ingenii) as a shared mark of prolonged heating and of iron that came from the Earth.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The quantitative values that define the two key observations (the distribution of the concentric belts and the balance of foreign material in the upper crust) and the Coriolis friction in the Martian ejection conduit are formulated as follows (Divisions I, K, L):

$$h_{\text{Model 1}}:\ 26.3\ \text{km (cap)} < 30.3\ \text{km (belt)} < 36.2\ \text{km (far)} \quad (\Delta h = 9.9$$

$$M_{\text{foreign, measured}} \leq 0.9\%\text{–}2\% \ll M_{\text{foreign, asteroidal}} = 4.4\text{–}5.0 \times 10^{19}\ \text{kg}\ (2\%$$

These data illustrate how two independent tests — the geometry of crustal thickness and craters on the surface on the one hand, and the isotopic fingerprint of the metal in the crater melt on the other — set sharp, unambiguous conditions of verification and falsification for the whole model.

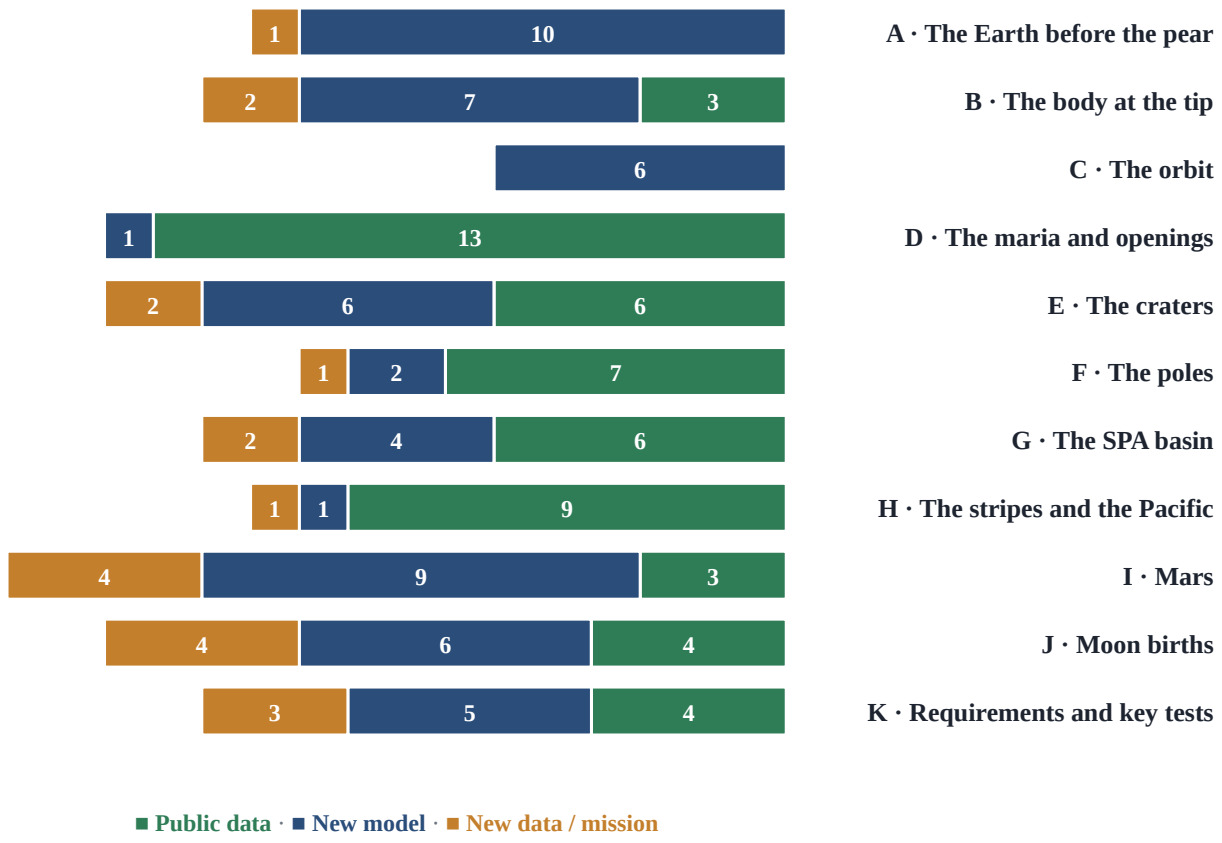
06 · THE SIGNATURE

The signature in reality: the complete road map of the 132 topics across the whole LEF series

Paper 14 binds all 14 papers of the series (Zenodo DOI: 10.5281/zenodo.22743615) into a multidisciplinary research system open to the whole planetary-science community:

- **Immediate work for data researchers (55 topics):** analysis of the GRAIL and GRGM1200B grids, the Robbins crater catalogue, thorium and titanium maps, GEBCO bathymetry of the Pacific and the shape models of Phobos and Deimos — with no need for supercomputing resources or new samples.

- **Targets for the space missions of the coming decade (20 topics):** analysis of norites from the Chang'e-6 samples (topic 123), high-pressure minerals in the Apollo 14 samples (topic 22), the Japanese MMX mission to Phobos in 2027–2031 (topics 98–104), and future missions to Uranus and the Kuiper belt (topics 112, 115, 117, 119).



All 132 research topics by division and level of readiness: 55 on public data, 57 require a new model, 20 require new data or a mission.

The lab box: an overall summary of all 11 divisions (A–K) and the 132 research topics of the LEF model (Paper 14)

Division	Field of research	Topics	Public data	New model	New data / mission
A	The Earth before and during the pear	1–11 (11)	0	10	1
B	The body at the tip — shape, rounding and release	12–23 (12)	3	7	2

C	The orbit and tidal evolution	24–29 (6)	0	6	0
D	The maria and the openings	30–43 (14)	**13**	1	0
E	The craters and their source (key observation 49)	44–57 (14)	6	6	2
F	The poles and the compaction of the rounding	58–67 (10)	7	2	1
G	The South Pole–Aitken basin (SPA)	68–79 (12)	6	4	2
H	**The stripes and the floor of the Pacific**	**80–90 (11)**	**9**	**1**	**1**
I	Mars and other bodies (the MMX mission)	91–106 (16)	3	**9**	**4**
J	The birth of moons in the solar system	107–120 (14)	4	6	**4**
K	**Requirements and key tests (121–123)**	**121–132 (12)**	**4**	**5**	**3**
Total	**All 11 divisions (A–K)**	**1–132 (132)**	**55 (41.7%)**	**57 (43.2%)**	**20 (15.1%)**

07 · NEXT

The door ahead — the end of the whole series

In Chapter 28 we completed the full survey of **all 14 papers of the LEF (Lunar Elevation and Fission) series** over 28 ordered chapters: from the introduction of principle (Paper 1), through the physics of the corotation line, the Earth's pear, the orbit, the maria, the craters, the poles and the South Pole–Aitken basin (Papers 2–8), to the stripes in the Pacific, the moons of Mars, the law of moons and the rings across the solar system (Papers 9–12), and up to the quantitative requirements and the map of 132 future research topics (Papers 13–14).

The complete 28-chapter book is now entirely at your disposal — both as a structured study guide and as an open scientific road map for further

investigation, simulation and empirical testing of the secrets of the Earth–Moon system and of the whole solar system.

← Chapter 27 · Prologue

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